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Non-Adaptive Scheduling of Entanglement Purification for All-Photonic Quantum Repeaters

Research Internship Defense — Master 2 Quantum, Mathematics and Informatics (QMI)

Eliott Flechtner

EPITA & Institut Polytechnique de Paris | AQUA Lab, Keio University, Japan

SUPERVISORY COMMITTEE

Prof. Rodney Van Meter (*Keio University*)
Proj. Assoc. Prof. Michal Hajdušek (*Keio University*)
Proj. Assist. Prof. Naphan Benchasattabuse (*Keio University*)

EXAMINATION COMMITTEE

Prof. Romain Alléaume (*Télécom Paris, IP Paris*)
Prof. Daniel E. Browne (*University College London*)

Why Quantum Repeaters

- Fiber loss grows exponentially with distance
- Direct transmission cannot scale
- Repeaters break the channel into shorter, purified hops
- 1G/2G: memory-based, limited by dephasing & two-way signaling
- 3G, all-photonic: flying graph states, loss tolerance via tree encoding

Fiber loss vs. all-photonic repeater concept

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Half-RGS Repeaters

- HRGS: half of a repeater graph state, held at one station
- End nodes act as their own entanglement source
- Memory cost grows additively with hop count, not multiplicatively
- Outer photons meet at analyzer stations
- Linear-optical BSM success capped at 50% → redundant arms

HRGS combination and N -hop chain

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The Scheduling Gap

- Purification can be inserted at generation, link, intermediate span, or end-to-end
- Circuit, copy count and order can vary between stages
- Prior work (Integrating paper) shows this flexibility with one hand-picked schedule
- Which allocation is actually best under a fixed budget was left open

Purification insertion points across spans

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Noise Model and Purification Circuits

- Bell-diagonal states, error vector $e = (w, x, y, z)$, fidelity $F = w$
- Inner-qubit errors bias x, z but leave $y \approx 0$
- Three 2-to-1 circuits: C_{ZX}, C_{XZ}, C_{YY}
- C_{YY} used first to remove the single-sided bias

Circuit layouts

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Representing a Partial Resource

Every intermediate resource is a state:

$$S = (e, s, t, t_{\text{gen}}, \kappa, r, h)$$

- e : error vector, s : Pauli frame bit
- $t - t_{\text{gen}}$: idle time $\rightarrow$ memory dephasing
- κ : span, r : purification rounds, h : heralding status

Span-matching rule

Two states purify together only if $\kappa(S_1) = \kappa(S_2)$ — construction history doesn't matter.

A Schedule Is a DAG

$$\Sigma = (T, \phi), \phi : T \rightarrow \mathcal{K}$$

- Gen, Join, Swap, Purify, Idle, Herald, PauliCorrect
- Optimistic: heralding deferred to a single node at the root

$N = 2$ worked example DAG

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Evaluating a Schedule

- Single bottom-up pass, $O(|T|)$: fidelity, cost, latency
- Rate accounts for full-schedule restart on failure:

$$R(\Sigma) = \frac{P_{\text{success}}}{T_{\text{attempt}}}$$

$$\Sigma^* = \arg \max_{\Sigma \text{ feasible}} R(\Sigma) \text{ s.t. } F(\Sigma) \geq F_{\min}$$

- Concurrency $M(\Sigma)$: min. open branches, via Sethi–Ullman recurrence
- Exact register bound, no timing info needed

Searching the Schedule Space

- Combinatorial explosion: circuit choice, split points, copy counts
- Exhaustive search intractable beyond $N = 3$

Three tiers:

1. Fixed baselines (raw, uniform link-level, paper schedule)
2. Exact span-partition dynamic program
3. Bounded beam search with same-span pumping

Baselines → exact DP → beam search

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Exact Search: Span-Partition DP

- Memoize best sub-schedules per span, reuse to build wider spans:

$$\text{Span}(a, c) = \text{Span}(a, b) \bowtie \text{Span}(b, c)$$

- Keep a Pareto set over (cost, fidelity, success prob.)
- Even a locally worse candidate can win later once swapped

Span memoization & Pareto pruning

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Pumping and Beam Search

- Same-span pumping: purify against a fresh copy of the same span, not just a different one
- Beam width 25 keeps frontiers tractable (uncapped: 1000+ candidates)
- Half the slots for top fidelity, half for best rate among feasible candidates

Beam frontier retention strategy

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Validating the Evaluator

- Reproduces the Integrating paper's Fig. 5 fidelity curves
- Raw chain, heralded pumping, optimistic pumping
- Match to four significant figures

Schedule	Ref.	Ours
Raw	0.823	0.8234
Heralded	0.917	0.9168
Optimistic	0.929	0.9295

Fidelity reproduction curve

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Beating the Paper's Schedule

Reference config: $N = 10$, $e_d = 0.01$, $F_{\min} = 0.90$, $E_{\max} = 100$

Schedule	C	F	Rate	Rel.	
Paper schedule	100	0.93	4,055.92	$1.00\times$	hand-picked
Matched cost	100	0.92	4,158.14	+2.5%	same budget
Budget-relaxed	60	0.91	6,195.95	+52.8%	60% of cost

- The paper schedule over-purifies everywhere uniformly
- Search spends the budget where it's actually needed

The Real Finding: Feasibility, Not Just Speed

- Turn on inner-qubit error ($p_{\text{in}} = 3 \times 10^{-3}$)
- Paper schedule drops below F_{min} : $F = 0.67$ (18 arms), $F = 0.87$ (6 arms)
- Search finds a feasible schedule at the **same cost**
- Same pattern on 100 randomized heterogeneous networks: mean +20% rate, never worse
- One weak hop with $12\times$ noise: uniform purification fails, search concentrates effort there

Fixed schedule vs. feasible searched schedule

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Scaling with Chain Length and Timing

- Minimum budget to hit $F \geq 0.9$, for $N = 10$ to 28
- Fit: $E_{\max}^{\min} \approx 1.19 N^{1.67}$ — superlinear
- Paper's linear $10N$ rule breaks around $N = 28$
- Heralded pumping: waiting copies decohere, F drops $0.92 \rightarrow 0.25$
- Optimistic schedules avoid the wait, stay flat

Budget scaling vs. dephasing decay

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When Does Scheduling Actually Matter

- Low noise ($e_d \leq 0.005$): easy, everything clears the floor
- High noise ($e_d \geq 0.016$): nothing clears the floor
- In between: **scheduling is decisive**
- Pareto quality plateaus past $C \approx 60$

Scope

Beam search is heuristic — results are existence proofs within the searched families, not global optima.

Pareto trade-off frontiers

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Conclusion

- DAG-based schedule model + validated evaluator
- Three-tier search: baselines, exact DP, beam search
- Main result: scheduling buys **feasibility**, not just a rate bump
- Budget scales roughly as $N^{1.67}$
- Next: adaptive (MDP-based) scheduling, mesh topologies

Code

[github.com/ElliottFlehtner/
entanglement-purification-scheduler](https://github.com/ElliottFlehtner/entanglement-purification-scheduler)

Thank you.

Questions welcome.

Backup: Sethi–Ullman Concurrency Algorithm

- Bottom-up register assignment
- Evaluate the heavier branch first, hold result in one register
- Time $O(|T|)$, space $O(h)$

RegisterRequirement(v) pseudocode

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Backup: DP vs. Beam Search Runtime

- Exact DP explodes beyond $N = 10$
- $N = 18$: DP times out (300s); beam search finds $R = 1,214$ in 153.5s
- Beam search scales predictably up to $N = 28$

Runtime and score benchmarks

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